WASP-36b

R-0988

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-36_180407 · created 2026-07-29T22:40:23+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2458215.6394 +/- 0.0028 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.136 +/- 0.045
Transit depth (Rp/Rs)^2	1.85 +/- 1.22 [%]
Orbital Inclination (inc)	86.61 +/- 2.68
Airmass coefficient 1 (a1)	1.033 +/- 0.019
Airmass coefficient 2 (a2)	-0.024 +/- 0.053
Scatter in the residuals of the lightcurve fit is	1.81 %
Best Comparison Star	#1 - [195, 278]
Optimal Aperture	3.1
Optimal Annulus	4.28
Transit Duration (day)	0.0417 +/- 0.0063

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.136 ± 0.045 vs 0.1368 (deviation 0.02σ)
Duration fit vs published	0.0417 d vs 0.0758 d (deviation 5.42σ)
Mid-time O–C	-6.8 min
Depth SNR	3.0
Residual scatter	1.81 %

Light curve

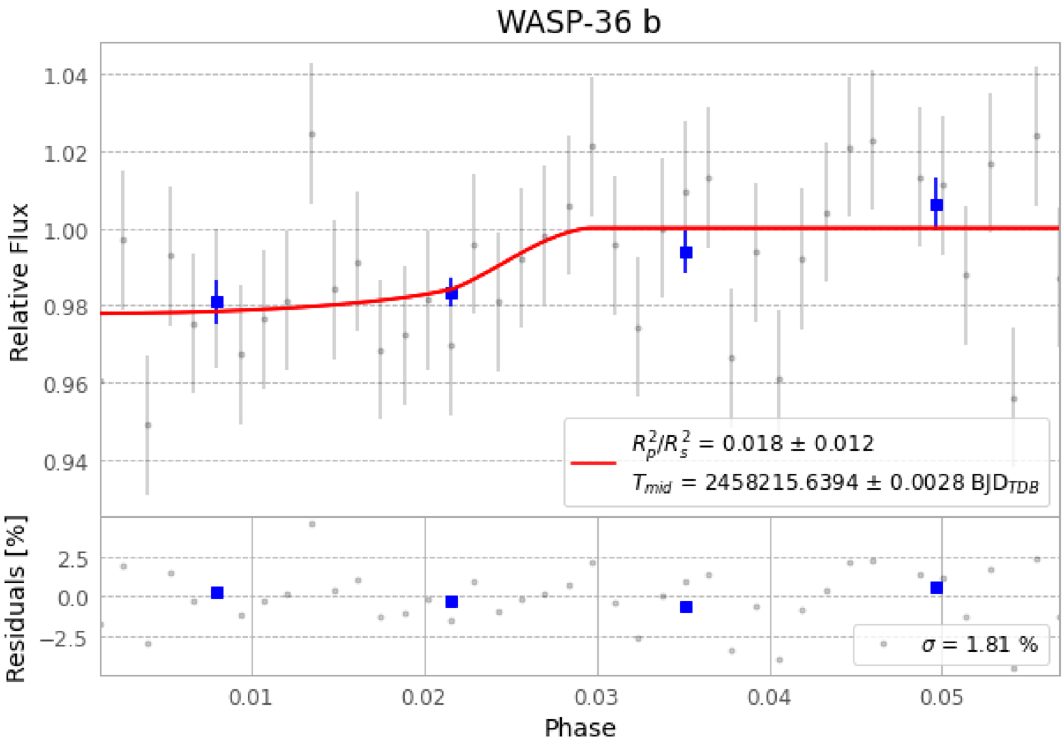

Figure 1 — FinalLightCurve_WASP-36 b_2018-04-06.png (SHA-256 94fc39ad7643be28…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [338, 259], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 a61cd328d750ec43…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9`

Data provenance

42 FITS frames (28 MB), `WASP-36180407031812.FITS` → `WASP-36180407052112.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-36-180407.log (2bfcf3c99596…), FinalLightCurve_WASP-36 b_2018-04-06.png (94fc39ad7643…), FinalLightCurve_WASP-36 b_2018-04-06.csv (2aa370c0a8c3…)

Compute resources

Wall time 420.9 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)